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Roll No

EE-6002-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Power Electronics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Draw and explain the two transistor model of SCR and derive an expression for anode current. 7
b) Explain the different methods for turning on of SCR. Briefly discuss the Resistance, capacitance firing circuit of SCR. 7
2. a) Explain the operation of single phase full controlled bridge rectifier with R-load. Derive the expressions for average load voltage, load current and RMS value of output voltage. 7
b) Explain the operation of single phase half controlled bridge rectifier with R-load. Derive the expressions for average load voltage, load current and RMS value of output voltage. 7
3. a) Explain the operation of single phase bridge inverter with waveforms. 7
b) Distinguish between current source inverter and voltage source inverter. 7

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4. Explain the operation of voltage commutated thyristor chopper. Draw the necessary waveforms. 14
5.
 - a) Explain the operation of single phase AC voltage controller with R load. Derive the expressions For RMS load voltage and RMS load current. 7
 - b) Draw the basic circuit diagram of a Boost regulator and explain its working. Obtain an expression for output voltage. 7
6.
 - a) Draw and explain the static characteristics of SCR. 4
 - b) Define commutation SCR and explain the class-A commutation technique for SCR. 5
 - c) Explain the class-D commutation technique for SCR. 5
7.
 - a) Explain the operation of three phase half controlled bridge rectifier with R-load. 7
 - b) Define the conduction angle for a thyristor and discuss the effect of source inductance in the performance of single phase fully controlled full wave bridge rectifier. 7
8. A three phase fully controlled bridge rectifier is connected to 3-phase ac supply of 400V, 50Hz and operates at a firing angle $\alpha = 30^\circ$. The load current is maintained constant at 12A and load voltage is 330V. 14
Find:
 - i) Source inductance
 - ii) Load resistance
 - iii) Overlap angle

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